AUTOMATIC SMART IRRIGATION AND MONITORING SYSTEM

A project report submitted in partial fulfilment of requirements to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

For the award of the degree of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

Under the esteemed guidance of

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DEPARTMENT OF

ELECTRONICS & COMMUNICATION ENGINEERING

SWARNANDHRA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Accredited by NAAC with 'A' Grade-3.02/4.00 CGPA and Accredited by NBA

Approved by A.I.C.T.E & Permanently Affiliated to JNTU Kakinada

Seetharampuram, Narsapur-534 280, West Godavari Dt., A.P

2022-2026

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BONAFIDE CERTIFICATE

This is to certify that this project report "**AUTOMATIC SMART IRRIGATION AND MONITORING SYSTEM**" is the bonafide work of M.LAKSHMI PARVATHI (22A21A0469), K.V.S.S.AVINASH (22A21A04B6), A.SOWMYA (22A21A0470), K.TARUN (22A21A04A1), K. SOMASEKHAR (20A21A0486), in partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in **Electronics and Communication Engineering, JNTUK**, Kakinada of during the academic year **2025-2026.**

PROJECT GUIDE	**HEAD OF THE DEPARTMENT**
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EXTERNAL EXAMINE

ACKNOWLEDGEMENT

A project is a team work that is supported by a large number of well-wishers without any exception to this, we would like to thank a few people who have helped us to convert our dream to reality.

We extend our heartfelt gratitude to the almighty with profound gratitude, respect and pride, we express our sincere thanks to the management of Swarnandhra College of Engineering & Technology **Sri K. V Satyanarayana, Chairman, Sri. K. V. Swamy, Treasurer, and Sri. A. Srihari, Director** for providing necessary arrangements to carry out of this project.

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DECLARATION

we certify that

a. The Major project work has been done under the guidance of my supervisor.

b. The work has not been submitted to any other university for the award of any degree.

c. The guidelines of the university are followed in writing the thesis.

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ABSTRACT

The Automatic Smart Irrigation and Monitoring System is designed to optimize water usage in agriculture by automating the irrigation process based on real-time soil conditions. This system uses a soil moisture sensor to detect the water content in the soil and an ESP32 microcontroller to process the data. when the soil moisture level falls below a set threshold, the system automatically activates a DC submersible pump through a relay to supply water to the plants. An LCD display is used to show real-time parameters such as soil moisture levels and system status. The system operates without the need for internet connectivity, making it cost-effective and suitable for rural areas. By reducing manual intervention and preventing over-irrigation, the system helps conserve water, improve crop yield, and promote efficient farming practices.

CHAPTER-1
INTRODUCTION

1.1 INTRODUCTION TO EMBEDDED SYSTEM

An embedded system is a special-purpose computer system designed to perform one or a few dedicated functions, sometimes with real-time computing constraints. It is usually embedded as part of a complete device including hardware and mechanical parts. In contrast, a general-purpose computer, such as a personal computer, can do many different tasks depending on programming. Embedded systems have become very important today as they control many of the common devices we use.

Since the embedded system is dedicated to specific tasks, design engineers can optimize it, reducing the size and cost of the product, or increasing the reliability and performance. Some embedded systems are mass-produced, benefiting from economies of scale.

Physically embedded systems range from portable devices such as digital watches and MP3 players, to large stationary installations like traffic lights, factory controllers, or the systems controlling nuclear power plants. Complexity varies from low, with a single microcontroller chip, to very high with multiple units, peripherals and networks mounted inside large chassis or enclosure.

In general, "embedded system" is not an exactly defined term, as many systems have some element of programmability. For example, Handheld computers share some elements with embedded systems such as the operating systems and microprocessors which power them but are not truly embedded systems, because they allow different applications to be load and peripherals to be connected.

An embedded system is some combination of computer hardware and software, either fixed in capability or programmable, that is specifically designed for a particular kind of application device. Industrial machines, automobiles, medical equipment, cameras, household appliances, airplanes, vending machines, and toys (as well as the more obvious cellular phone and PDA) are among the myriad possible hosts of an embedded system. Embedded systems that are programmable are provided with a programming interface, and embedded systems programming is a specialized occupation. Certain operating system language platforms are tailored for the embedded market, such as Embedded Java and Windows XP Embedded.

1.2 CHARACTERISTICS OF EMBEDDED SYSTEMS

- **Speed (bytes/sec):** Should be high speed
- **Power (watts):** Low power dissipation
- **Size and weight**: As far as possible small in size and low weight
- **Accuracy (%error):** Must be very accurate
- **Adaptability:** High adaptability and accessibility
- **Reliability:** Must be reliable over a long period of time

1.3 APPLICATIONS OF EMBEDDED SYSTEMS

We are living in the Embedded World. You are surrounded with many embedded products and your daily life largely depends on the proper functioning of these gadgets. Television Radio player of your living room, Washing Machine or Microwave Oven in your kitchen, Card readers, Access Controllers, Palm devices of your work space enable you to many of your tasks very effectively

- **Robotics:** industrial robots, machine tools, Robocop soccer robots
- **Automotive:** cars, trucks, trains
- **Aviation:** airplanes, helicopters
- Home and building automation

CHAPTER-2
LITERATURE SURVEY

2.1 LITERATURE SURVEY

Organic farming doesn't just reduce toxic runoff in our water and air. Organic farms use soil-building and biodiversity practices that are more resilient to extreme weather conditions such as drought and hurricanes. Worldwide studies show that organic farms experience fewer losses of crops, topsoil, and income after extreme weather events compared to non-organic farms. Buying organic invests in the future of food. When organic farmers like Patrick Martin of Frantoio Grove treat their farmland like an ecosystem, they build high levels of soil organic matter. Healthy soil stores carbon and helps reduce levels of the greenhouse gas carbon dioxide (CO_2) in the atmosphere.[1]

"In the present era, the greatest problem faced by world is scarcity of water. Agriculture is an occupation demanding plenty of water. Irrigation refers to the supplying water to cultivation land as supplementation of rain fall. There are various types of irrigation system that have been adopted. The efficiency of the irrigation system in conserving water is not appreciable. Moreover, the water requirement by the crop depends on type of soil, crop and environmental parameters like temperature and humidity. Conventional Irrigation system either result in over irrigated or under irrigated land. As growth and development of plants is prevented due to scarcity of water, similarly excessive water has adverse effect on growth of plants. Under Conventional Irrigation system, many parts of irrigated field are over or under irrigated due variability in the water holding capacity of land, water infiltration and water runoff. Over irrigated area, suffers from poor plant health due to increase in salinity. Excessive water replaces air in pores of the soil. Hence roots of the plants do not get sufficient air. It may lead to leaching. While under irrigated area suffers water stress. Hence efficient water management plays key role in agriculture."[2]

Agriculture consumes an important ratio of the water reserve in irrigated areas. The improvement of irrigation is becoming essential to reduce this high water consumption by adapting supplies to the crop needs and avoiding losses. This global issue has prompted many scientists to reflect on sustainable solutions using innovative technologies, namely Unmanned Aerial Vehicles (UAV), Machine Learning (ML), and the Internet of Things (IoT). [3]

Overview of the use of these new technologies in the analysis of the water status of crops for better irrigation management, with an emphasis on arboriculture. The review demonstrated the importance of UAV-ML-IoT technologies. This contribution is due to the relevant information that can be collected from IoT sensors and extracted from UAV images through various sensors (RGB, multispectral, hyperspectral, thermal), and the ability of ML models to monitor and predict

water status. The review in this paper is organized into four main sections: the use of UAV in arboriculture, UAV for irrigation management in arboriculture, IoT systems and irrigation management, and ML for data processing and decision-making. A discussion is presented regarding the prospects for smart irrigation using geospatial technologies and machine learning.[4]

Drip irrigation is widely acknowledged for its cost-effectiveness and efficiency in contrast to traditional methods that rely heavily on human oversight. This research paper underscores the potential to enhance water utilization efficiency while maintaining optimal yields through the implementation of sound water management strategies and the integration of cutting-edge technologies. The study introduces an automated irrigation control system designed to autonomously supply adequate water to crops across all agricultural seasons. This system is intended to create an automated irrigation mechanism which turns the pumping motor ON and OFF on detecting the moisture content of the earth. Utilizing inexpensive sensors and straightforward circuitry renders this project a cost-effective solution, making it accessible even to economically disadvantaged farmers. [5]

The integration of IoT in drip irrigation systems is crucial for enhancing agricultural sustainability and efficiency, particularly in the context of climate change and food security challenges. This review paper synthesizes findings from 56 studies, highlighting that IoT-enabled systems can achieve significant water savings of up to 70% and yield increases of 30% through advanced machine learning models, while also identifying key barriers to widespread adoption such as high costs and technical complexity.[6]

Irrigation System powered by Internet of Things (IoT) technology, aimed at automating the irrigation process using real-time soil moisture and weather data. The system leverages a combination of sensors, microcontrollers, cloud services, and mobile applications to monitor and control irrigation activities remotely. It ensures efficient water management, reduces manual intervention, and improves crop yield. The proposed system demonstrates the potential of IoT in transforming traditional farming into a data-driven, eco-friendly process.[7]

The lack of fresh water is a rising concern, particularly in the Mediterranean countries or southern Asian countries such as India. Among the countries in Europe, the Mediterranean countries are the most vulnerable to drought. A connection has been established between climate policies and water management. Water management can be affected by different variables such as the water demand from the different sectors or the consequences of some degrees of warming on hydrological resources. Climate change and its effects are a recurrent topic in research papers regarding water resources and agriculture. The possible consequences of global warming have led to the consideration of creating water adaptation measures to ensure the availability of water for food production and people and to maintain ecosystems. Furthermore, the safety of the water to be consumed by humans and to be returned to the environment must be ensured. The possible risks of climate change are an increase in water shortage, the reduction of water quality, the increase in water and soil salinity, the biodiversity loss, the increase in irrigation requirements or the possible cost of emergency and remediation actions. These reasons have led to an increase in the number of studies focused on reducing water usage in irrigation processes. Some of these studies suggest.[8]

Automation of farm activities can transform agricultural domain from being manual and static to intelligent and dynamic leading to higher production with lesser human supervision. This paper proposes an automated irrigation system which monitors and maintains the desired soil moisture content via automatic watering. Microcontroller ATMEGA328P on Arduino uno platform is used to implement the control unit. The setup uses soil moisture sensors which measure the exact moisture level in soil. This value enables the system to use appropriate [9]

Quantity of water which avoids over/under irrigation. IOT is used to keep the farmers updated about the status of sprinklers. Information from the sensors is regularly updated on a webpage using GSM-GPRS SIM900A modem through which a farmer can check whether the water sprinklers are ON/OFF at any given time. Also, the sensor readings are transmitted to a Thing speak channel to generate graphs for analysis.[10]

Inline Drip Irrigation Systems: Designed to deliver water directly to plant roots, reducing water waste, evaporation, and runoff. Mini Sprinkler Irrigation Systems:

Specializes in rain-like irrigation methods for efficient water distribution. Smart Irrigation Equipment: Offers accessories such as rain pipe cocks, lateral cocks, and various filter types for automated systems. Automatic Drip Kits: Provides Kits for gardens, lawns, and small farms that can be set for automatic, timed, or controlled watering. Balaji Polymers. Balaji Polymers System Features & Benefits, Water Efficiency: Minimizes evaporation and eliminates water waste, aimed at high productivity and maximum yield. High Performance Components: Uses high-quality PVC/steel materials with durable, UV-resistant, and corrosion-resistant components. Customization: Offers customizable pressure (1-3 bar), flow rates, and layout designs, including 360-degree adjustable sprinklers.[11]

Irrigation is one of the most powerful sources in India but it is hard for an individual person to monitor continuously and regularly. This is due to laziness of mankind. In order to make this irrigation easier our system comprises some changes in the usual irrigation system. The newly developed project controls water supply automatically in water crisis areas through moisture sensor. This paper covers the application of Sensor based Irrigation system through wireless sensor networks, which uses a renewable energy as a source. In this system Wireless Sensor Networks Plays a major role in Environment monitoring system and provides unmanned irrigation. WSN consists of moisture sensors, Energy harvesting systems, embedded controllers and uses Super capacitors as storage device. To develop a reliable system for smart irrigation of greenhouses using artificial neural networks, and an IoT architecture. Our solution uses four sensors in different layers of soil to predict future moisture. Using a data set, we collected by running experiments on different soils, we show high performance of neural.[12]

Networks compared to existing alternative method of support vector regression. To reduce the processing power of neural network for the IoT edge devices, we propose using transfer learning. Transfer learning also speeds up training performance with small amount of training data, and allows integrating climate sensors to a pre-trained model, which are the other two challenges of smart irrigation of greenhouses. Our proposed IoT architecture shows a complete solution for smart irrigation.[13]

Strawberries are widely appreciated for their aroma, bright red colour, juicy texture, and sweetness. They are, however, among the most sensitive fruits when it comes to the quality of the end product. The recent commercial trends show a rising number of farmers who directly sell their products and are interested in using smart solutions for a continuous quality control of the product. Cloud-based approaches for smart irrigation have been widely used in the recent years. However, the network traffic, security and regulatory challenges, which come hand in hand with sharing the crop data with third parties outside the edge of the network, lead strawberry farmers and data owners to rely on global clouds and potentially lose control over their data, which are usually transferred to third party data centers. In this paper, we follow a three-step methodological \implement and validate a solution for smart strawberry irrigation in greenhouses, while keeping the corresponding data at the edge of the network.[14]

The increasing global demand for sustainable agriculture necessitates intelligent monitoring systems that optimize resource utilization and plant health management. Traditional farming methods rely on manual observation and periodic watering, often leading to water wastage, inconsistent plant growth, and delayed response to environmental changes. This paper presents a comprehensive IoT-based smart plant monitoring system that integrates multiple environmental sensors with automated irrigation and cloud analytics. The proposed system utilizes an ESP32 microcontroller to collect real-time data from DHT22 (temperature/humidity), HC-SR04 (water level), and soil moisture sensors, with visual feedback through an OLED display and auditory alerts via a buzzer. All sensor data is wirelessly transmitted to the ThingSpeak cloud platform for remote monitoring, historical analysis, and automated alert generation. Agricultural practices need to be transformed in order to overcome future food scarcity due to overpopulation across the globe. By employing emerging, disruptive technologies like IoT in the agricultural sector, it is possible to monitor farm fields using low-cost and low-power consuming devices, to automate irrigation systems for efficient usage of water resources. Weather forecast using IoT can help to plan farm filed activities like sowing, harrowing, harvesting, etc. This reduces negative impacts like yield losses due to uncertain weather.[15]

2.2 EXISTING SYSTEM

A traditional irrigation system works by manually supplying water to agricultural fields based on the farmer’s observation and experience. The farmer checks the soil condition physically and decides when to turn ON or OFF the water pump. Water flow is controlled using manual valves and switches without the use of sensors or lack of precise monitoring. As a result, water resources are wasted, crop growth may be affected, and continuous human effort is required to manage the irrigation process efficiently.automation. This method often leads to over-irrigation or under-irrigation due to

Soil Observation:

- The farmer manually inspects the soil condition to determine whether irrigation is required.

Decision Making:

- Based on experience, the farmer decides when to start or stop watering the field**.**

Manual Pump Operation:

- The water pump is switched ON manually to supply water to the crops.

Water Flow Control:

- Water flow is controlled using manual valves without any measurement monitoring.

Irrigation Process:

- Water is supplied continuously until the farmer decides to stop it.

Monitoring:

- The farmer periodically checks the field to ensure proper irrigation.

Stopping the Pump: Once sufficient watering is achieved, the pump is manually turned.

2.3 EXISTING SYSTEM BLOCK DIAGRAM

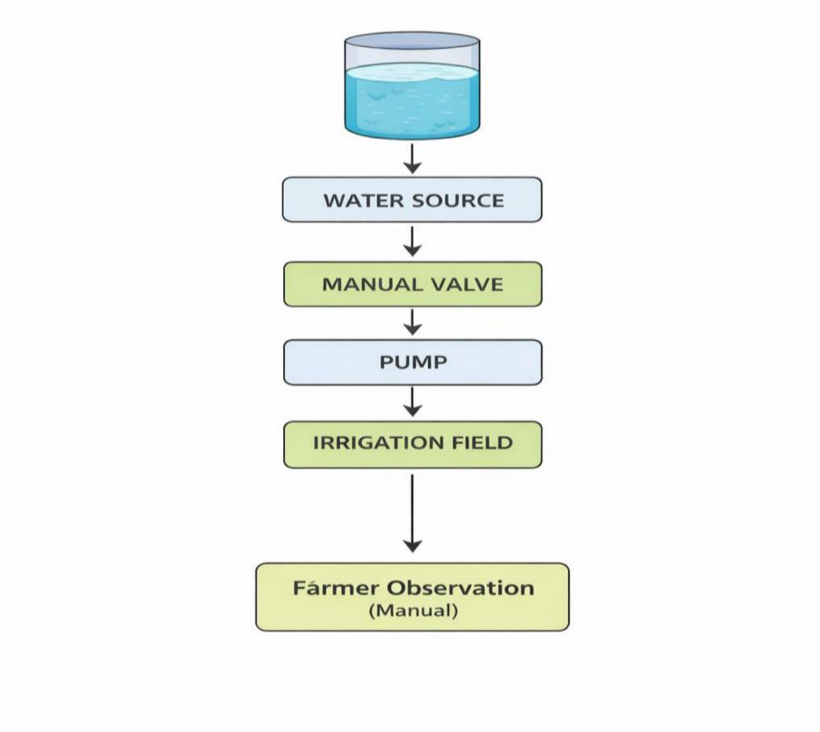

FIG. 2.1 EXISTING SYSTEM BLOCK DIAGRAM

FIG: 2.2 TRADITIONAL WATER IRRIGATION SYSTEM

2.4 LIMITATIONS

1. **Time Consuming Process**

   Manual inspection requires workers to walk long distances, making the process slow.

2. **Human Error**

   Detection depends on visual judgment, which may lead to missed cracks.

3. **Limited Crack Detection**

   Small, hidden, or internal cracks cannot be identified easily.

4. **Low Inspection Frequency**

   Tracks cannot be inspected regularly due to manpower limitations.

5. **High Labor Cost**

   Requires more workforce, increasing operational expenses.

6. **Safety Risks**

   Workers face danger from running trains and environmental conditions.

7. **No Real-Time Monitoring**

   There is no instant alert or live tracking system.

8. **Inaccurate Location Reporting**

   Exact crack location is difficult to record manually.

CHAPTER-3
PROPOSED PROJECT WORK

3.1 INTRODUCTION TO THE PROJECT

The concept of agriculture – the cultivation of crops for human survival – has been an essential part of civilization since ancient times. Over the years, traditional farming methods have evolved with technological advancements to improve productivity and efficiency. In the modern era, the integration of electronics and automation has opened new possibilities in agricultural practices. The emergence of smart systems and embedded technologies has enabled the development of intelligent solutions to address challenges such as water scarcity, labour shortage, and inefficient irrigation methods.

Efficient water management is one of the most critical aspects of agriculture, especially in regions facing irregular rainfall and limited water resources. Conventional irrigation systems rely heavily on manual operation, where farmers irrigate fields based on observation and experience. This often leads to over-irrigation or under-irrigation, resulting in water wastage, reduced crop yield, and increased operational costs. Additionally, continuous manual monitoring of soil and environmental conditions is time-consuming and not always accurate.

To overcome these limitations, the concept of an automatic smart irrigation and monitoring system has been introduced. This system utilizes sensors to monitor key parameters such as soil moisture and water flow, and automatically controls irrigation based on real-time data. By embedding intelligence into the irrigation process, the system ensures that crops receive the right amount of water at the right time, thereby optimizing water usage and improving crop health.

The integration of microcontrollers such as ESP32, along with sensors like soil moisture sensors and water flow sensors, enables efficient data processing and control. The system can automatically activate or deactivate a water pump using a relay module, based on the moisture level of the soil. Additionally, display units such as LCD screens provide real-time information to the user, enhancing monitoring capabilities without the need for constant manual supervision.

This automated approach not only conserves water but also reduces human effort and increases agricultural productivity. By leveraging modern electronic components and intelligent control mechanisms, the automatic smart irrigation and monitoring system represents a significant step toward sustainable and precision agriculture.

3.2 OBJECTIVES

- To design an automatic irrigation system that waters plants based on soil moisture level.
- To reduce water wastage by supplying only the required amount of water.
- To monitor soil conditions using sensors like soil moisture sensor and water flow sensor.
- To control the water pump automatically using a relay and microcontroller (ESP32).
- To display real-time data such as moisture level and water usage on an LCD display.
- To minimize human effort in agricultural irrigation.
- To improve crop growth
- To design an automatic irrigation system that waters plants based on soil moisture level.
- To develop a low-cost and efficient smart farming solution maintaining proper soil moisture.
- To develop a low-cost and efficient smart farming solution.

3.3 BLOCK DIAGRAM

- **Power Supply**: Provides required power to ESP32, sensors, relay, and LCD display.
- **Soil Moisture Sensor:** Detects soil moisture level and sends data to ESP32.
- **Water Flow Sensor:** Measures water flow rate and gives feedback to ESP32.
- **ESP32 Microcontroller:** Processes sensor data and controls the system operation.
- **Relay Module:** Acts as a switch to control the water pump based on ESP32 signals.
- **Water Pump:** Pumps water to the field when relay is ON and stops when OFF.
- **LCD Display (16x2 I2C):** Displays moisture level, water flow, and pump status.
- **Overall Function**: Automates irrigation by turning the pump ON/OFF based on soil moisture, reducing water wastage.

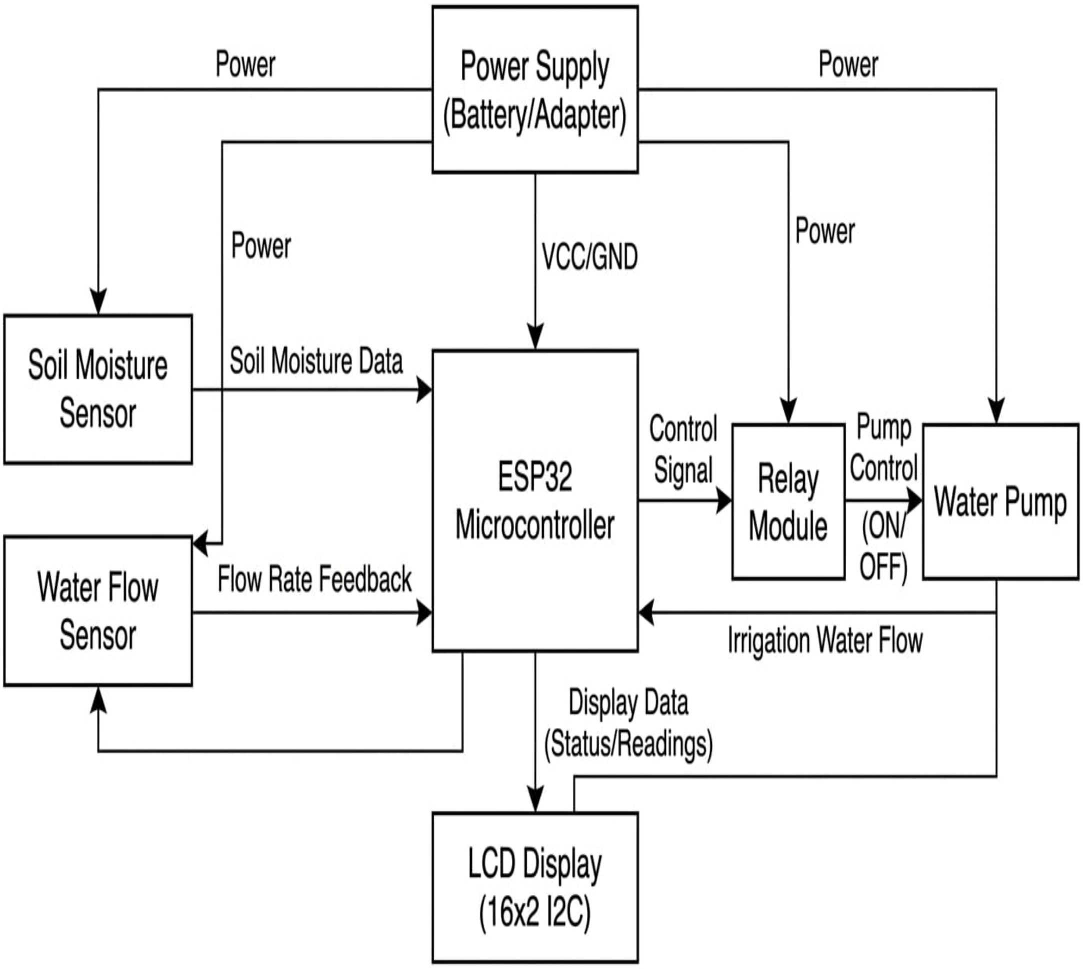

FIG 3.1 BLOCK DIAGRAM OF PROPOSED PROJECT WORK

- **Water Flow Sensor:** Measures water flow rate and gives feedback to ESP32.
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3.4 WORKING

The Automatic Smart Irrigation and Monitoring System works by continuously monitoring the soil moisture level and controlling the water supply accordingly. The system is powered by a suitable power source, and the soil moisture sensor detects the moisture content present in the soil. This data is sent to the ESP32 microcontroller, which compares the received value with a predefined threshold. If the soil is dry, the controller activates the relay module, which in turn switches ON the DC mini submersible pump to supply water to the field. When the soil reaches sufficient moisture, the controller automatically turns OFF the pump to prevent over-irrigation. Additionally, the water flow sensor measures the amount of water being supplied, and all the real-time information such as soil condition, pump status, and water usage is displayed on the LCD screen. Thus, the system ensures efficient water management, reduces human effort, and supports smart agricultural practices.

Key components and steps:

Sensor Setup:

- Soil moisture sensor and water flow sensor are placed in the field measure soil condition and water flow rate.

Data Collection:

- The ESP32 continuously reads data from the sensors such as moisture level and water flow.

Moisture Detection:

- The system checks whether the soil is dry or wet based on predefined threshold values.

Decision Making:

- If the soil moisture is below the set level, the ESP32 decides to start irrigation

Pump Activation:

- The relay module is activated, which turns ON the DC submersible Pump to supply water.

Water Flow Monitoring:

- The water flow sensor monitors the amount of water being supplied to avoid wastage.

Automatic Control:

- Once the soil reaches sufficient moisture level, the system automatically turns OFF the pump.

Display Output:

- The LCD display shows real-time values like soil moisture level and pump status.

Power Supply:

- The system is powered using lithium-ion batteries enursing continuous operation.

3.5 PROJECT KIT

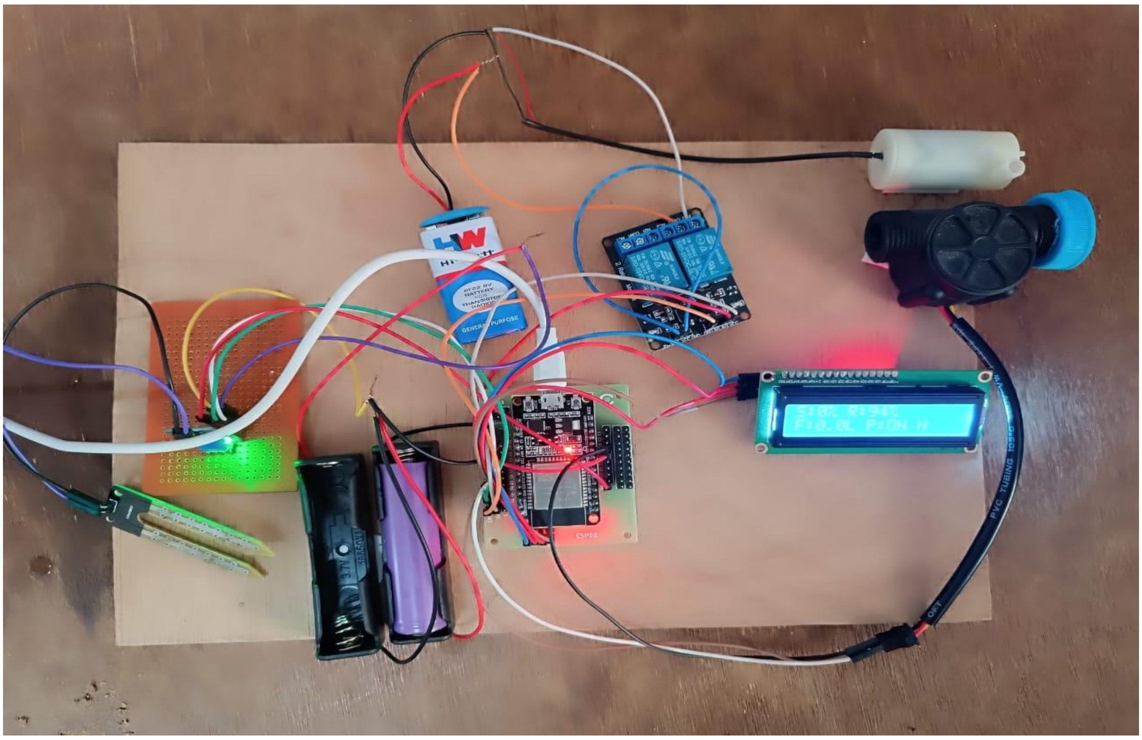

FIG 3.2 PROJECT KIT

CHAPTER-4
HARDWARE COMPONENTS

4.1 DESIGN HARDWARE:

(System Modules Set Up And Configuration)

- ESP32 Microcontroller
- Soil Moisture Sensor
- Water Flow Sensor
- Relay Module
- DC Mini Submersible Pump
- LCD Display (I2C)
- Lithium-ion Battery

4.1.1 ESP32 MICROCONTROLLER:

The ESP32 is a low-cost, low-power microcontroller with built-in Wi-Fi and Bluetooth capabilities. It is widely used in IoT applications due to its high performance, versatility, and energy efficiency. Developed by Espressif Systems, it supports dual-core processing and multiple peripheral interfaces.

Key Features and Specifications:

- **Processor**: Dual-core Tensilica Xtensa LX6 processor running up to 240MHz.
- **Memory:** Typically, 520 KB SRAM with external flash memory support (usually
  4MB or more).
- **Connectivity**: Built-in Wi-Fi (802.11 b/g/n) and Bluetooth (Classic + BLE).
- **GPIO Pins:** Around 30+ programmable GPIO pins supporting various functions.
- **Analog/Digital Features**: 12-bit ADC, DAC, PWM, touch sensors.
- **Communication Protocols**: Supports UART, SPI, I2C, CAN, and I2S.
- **Power Consumption**: Designed for low-power applications with deep sleep.

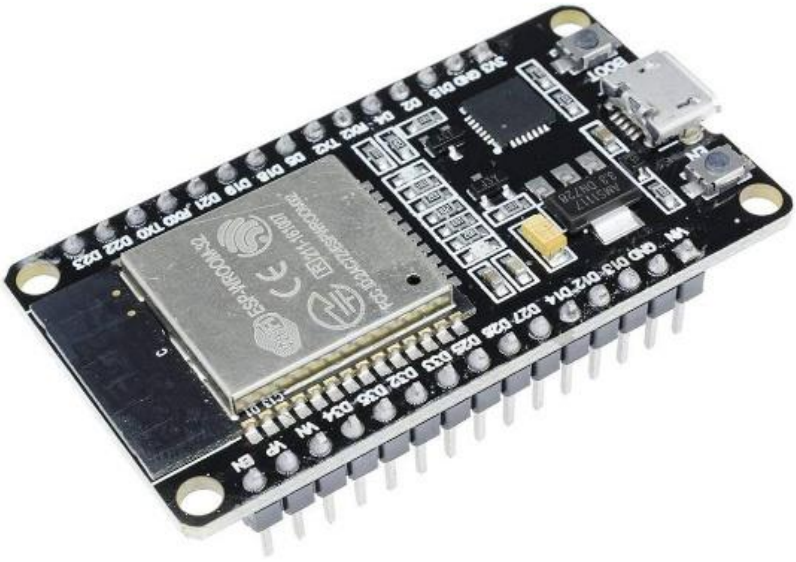

FIG 4.1: ESP32 MICROCONTROLLER

Advantages:

- **Performance**: High-speed processing with dual-core architecture, suitable for complex IoT applications.
- **Versatility**: Can be used in home automation, wearable devices, smart agriculture, and industrial IoT.
- **Wireless Capability**: Integrated Wi-Fi and Bluetooth eliminate the need for external modules.
- **Cost-Effectiveness**: Affordable compared to other microcontrollers with similar features.
- **Community Support**: Large developer community with extensive libraries and documentation.

4.1.2 SOIL MOISTURE SENSOR:

A Soil Moisture Sensor is a device used to measure the water content present in the soil. It works by detecting the electrical resistance or capacitance of the soil, which varies with moisture levels. These sensors are widely used in agriculture and IoT-based smart irrigation systems to monitor soil conditions and optimize water usage.

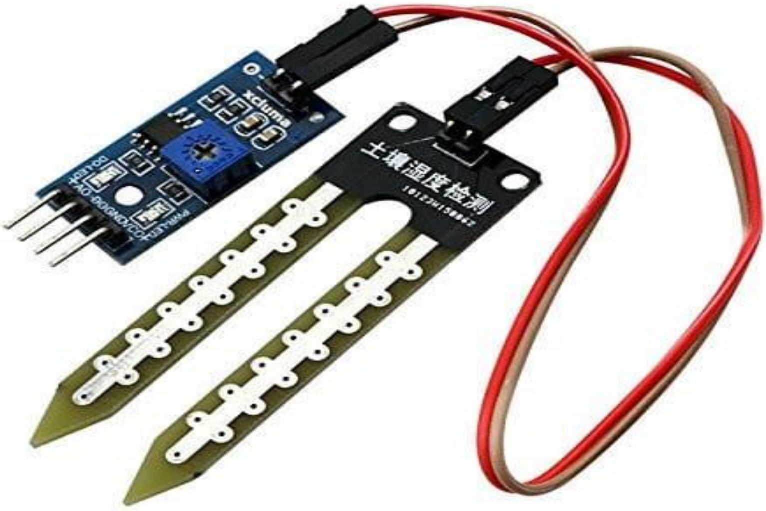

FIG 4.2 SOIL MOISTURE SENSOR

Key Features and Specifications:

- **Working Principle**: Measures soil moisture using resistance (conductivity) or capacitance method.
- **Output Type:** Provides analog and/or digital output signals.
- **Operating Voltage:** Typically operates at 3.3V to 5V.
- **Sensitivity Adjustment:** Comes with a potentiometer to adjust moisture sensitivity (in digital modules).
- **Probe Type:** Two conductive probes inserted into soil to detect moisture level.
- **Compatibility**: Easily interfaces with microcontrollers like Arduino, ESP32,Raspberry Pi.

Advantages

- **Efficient Water Usage:** Helps in automatic irrigation by supplying water only when needed.
- **Easy to Use:** Simple design and easy interfacing with controllers.
- **Low Cost**: Affordable and widely available sensor.
- **Real-Time Monitoring**: Provides continuous soil moisture data.
- **Versatility:** Suitable for agriculture, gardening, and smart farming applications.

4.1.3 WATER FLOW SENSOR:

A Water Flow Sensor is a device used to measure the rate of water flow through a pipe. It works using a rotating rotor and a Hall-effect sensor that generates pulses proportional to the flow of water. These sensors are commonly used in irrigation systems, water monitoring, and IoT-based applications to track water usage.

Key Features and Specifications:

- Working Principle: Uses a Hall-effect sensor to detect the rotation of an internal rotor caused by water flow.
- Output Type: Digital pulse output proportional to flow rate.
- Operating Voltage: Typically operates between 3.3V to 5V.
- Flow Range: Commonly measures flow rates from 1 to 30 liters per minute.
- Accuracy: Provides reasonably accurate flow measurement for general applications.
- Interface: Easily connects with microcontrollers like ESP32, Arduino, and Raspberry Pi.

FIG 4.3: WATER FLOW SENSOR

Advantages:

- Water Monitoring: Helps track water consumption in real-time.
- Automation Support: Useful in smart irrigation and
- automated water control systems.
- Easy Installation: Simple to install in pipelines.
- Low Cost: Affordable sensor for IoT projects.
- Reliable Operation: Provides consistent pulse output for flow calculation.

4.1.4 DUAL CHANNEL RELAY MODULE:

A Dual Channel Relay Module is an electronic switching device used to control two electrical appliances independently using a microcontroller. It allows low-voltage systems like ESP32 or Arduino to safely control high-voltage devices such as lights, motors, and pumps. Each channel operates as an electrically isolated switch.

Key Features and Specifications:

- **Channels:** Contains 2 independent relay channels for controlling two devices.
- **Operating Voltage:** Typically works at 5V (some modules support 3.3V logic).
- **Control Signal:** Can be triggered using digital signals from microcontrollers.
- **Relay Type:** Electromechanical relay with Normally Open (NO), Normally Closed (NC), and Common (COM) terminals.
- **Isolation**: Provides electrical isolation between control circuit and load circuit.
- **Load Capacity:** Can handle high voltage loads (e.g., 250V AC / 10A or 30V DC).

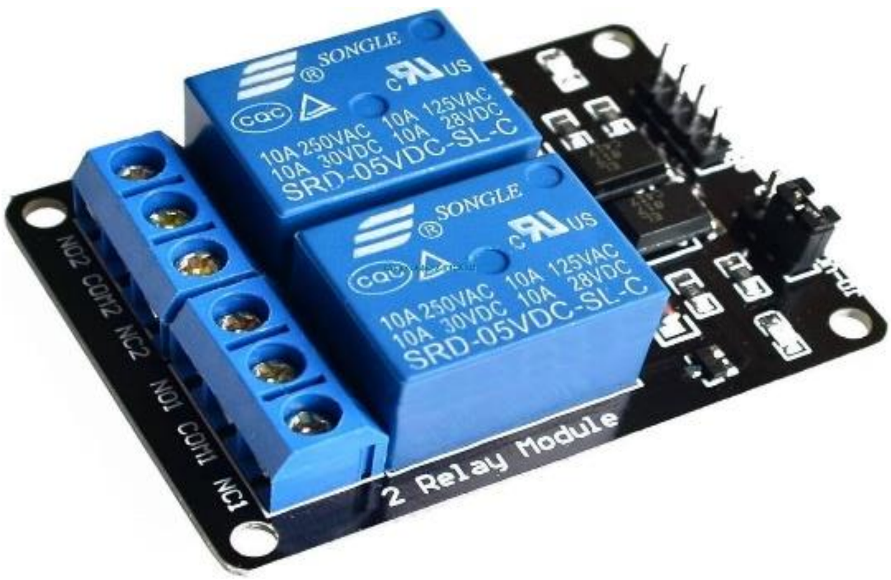

FIG 4.4: DUAL CHANNEL RELAY MODULE

Advantages:

- **Device Control**: Enables control of multiple appliances simultaneously.
- **Safety:** Protects low-power circuits from high voltage damage.
- **Versatility:** Used in home automation, industrial control, and IoT systems.
- **Ease of Use**: Simple interface with microcontrollers like ESP32 and Arduino.
- **Cost-Effective:** Affordable and widely available module.

4.1.5 DC MINI SUBMERSIBLE PUMP:

A DC Mini Submersible Pump is a small water pump designed to operate while fully submerged in water. It is commonly used in IoT and automation projects such as smart irrigation systems, water circulation, and aquarium setups. The pump runs on DC power and is controlled using microcontrollers through relay modules.

Key Features and Specifications:

- **Operating Voltage:** Typically operates at 3V to 12V DC (commonly 5V or 12V).
- **Working Principle:** Uses a small DC motor to drive an impeller that pushes water through the outlet.
- **Flow Rate:** Usually ranges from 80 to 120 litres per hour (varies by model).
- **Power Consumption:** Low power consumption suitable for battery-operated systems.
- **Size**: Compact and lightweight design for easy installation.
- **Material:** Made of waterproof and durable plastic body

-

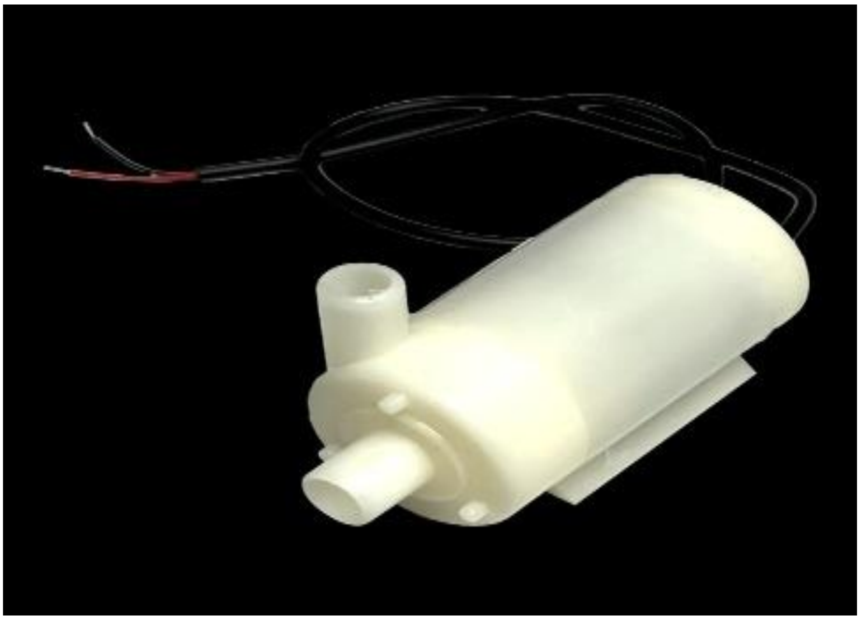

FIG 4.5: DC MINI SUBMERSIBLE PUMP

Advantages:

- **Efficient Water Pumping**: Suitable for small-scale water transfer applications.
- **Compact Design**: Easy to fit in small spaces like tanks or containers.
- **Low Power Usage**: Ideal for energy-efficient IoT systems.
- **Easy Integration:** Can be controlled using relay modules with ESP32 or Arduino.
- **Affordable**: Low-cost solution for water management projects.

4.1.6 LCD 16×2 (I2C) DISPLAY:

The LCD 16×2 (I2C) is a commonly used display module that can show 16 characters per line and has 2 lines. It is used to display data such as sensor readings, system status, and messages in embedded systems. The I2C interface reduces the number of pins required for connection, making it easier to use with microcontrollers like ESP32.

Key Features and Specifications:

- **Display Type:** 16×2-character LCD (16 columns and 2 rows).
- **Interface:** I2C communication (uses only 2 pins – SDA and SCL).
- **Operating Voltage:** Typically operates at 5V.
- **Controller**: Based on HD44780 LCD controller with I2C adapter.
- **Backlight:** Built-in LED backlight for clear visibility.
- **Address:** Default I2C address usually 0x27 or 0x3F (can vary).

FIG 4.6: LCD 16×2(I2C) DISPLAY

Advantages:

- **Less Pin Usage**: Uses only 2 communication pins instead of many GPIO pins.
- **Easy Interface**: Simple connection with ESP32, Arduino, and other controllers.
- **Clear Display:** Suitable for displaying text and basic information.
- **Cost-Effective**: Affordable and widely available.
- **Versatility:** Used in many projects like smart irrigation, home automation and monitoring system.

4.1.7 LITHIUM-ION BATTERY:

A Lithium-Ion Battery is a rechargeable energy storage device widely used in portable electronic devices and IoT systems. It stores electrical energy and supplies power to components like microcontrollers, sensors, and modules. These batteries are known for their high energy density, long life cycle, and lightweight design.

Key Features and Specifications:

- **Battery Type:** Rechargeable Lithium-Ion (Li-ion).
- **Nominal Voltage**: Typically 3.7V per cell.
- **Capacity:** Ranges from 1000mAh to 5000mAh or more depending on the model.
- **Recharge Cycles:** Can be recharged 300–500 times (or more).
- **Energy Density:** High energy storage compared to other batteries.
- **Charging Method:** Requires a proper charging circuit/module (like TP4056).

FIG 4.7: Lithium-Ion battery

Advantages:

- **High Efficiency:** Provides stable and reliable power output.
- **Lightweight:** Easy to use in portable and compact devices.
- **Low Self-Discharge:** Retains charge for a longer period when not in use.
- **Eco-Friendly:** More environmentally friendly compared to some traditional batteries.

4.1.8 HW BATTERY

The 9V HW (Hi-Watt) battery is a compact, non-rechargeable DC power source commonly used in electronic circuits. It provides a stable 9-volt output and is widely used in small embedded systems. In the smart irrigation and monitoring system, this battery can be used to power low-power components such as sensors, display modules, or as a backup power supply.

Key Features and Specifications:

- Type: 6F22 9V non-rechargeable battery
- Voltage: 9 Volts DC
- Chemistry: Zinc-carbon or alkaline
- Capacity: Typically, around 400–600mAh
- Terminals: Snap-type connector (positive and negative)
- Size: Compact and lightweight rectangular design
- Application: Suitable for low-power electronic device

Advantages

- Compact Size: Easy to fit into small circuit designs
- Ease of Use: Simple snap connector makes it easy to connect and replace.
- Availability: Easily available and low cost.

FIG-4.8 HW BATTERY

CHAPTER-5
WORKING FLOW

5.1 WORKING FLOW STEP BY STEP

5.1.1 SYSTEM INITIALIZATION

- The ESP32 microcontroller is powered ON.
- All sensors (soil moisture, water flow) and modules (relay, LCD) are initialized.

5.1.2 SOIL MOISTURE SENSING

- The soil moisture sensor continuously measures the water content in the soil.
- It sends analog signals to the ESP32.

5.1.3 WATER FLOW MONITORING

- The water flow sensor detects the rate of water flowing through the pipe.
- It sends pulse signals to the ESP32 for calculation.

5.1.4 DATA PROCESSING

- ESP32 reads sensor values using ADC and GPIO pins.
- The moisture value is compared with a predefined threshold level.

5.1.5 DECISION MAKING

- If soil moisture is below threshold → irrigation is required.
- If soil moisture is above threshold → irrigation is not required.
- Pump Control (Automation)
- When irrigation is needed, ESP32 activates the relay.
- The relay turns ON the DC submersible pump to supply water.
- When sufficient moisture is reached, the pump is turned OFF automatically.
- Real-Time Monitoring
- LCD display shows:
- Soil moisture level

5.1.6 WATER FLOW STATUS:

- Pump ON/OFF condition
- Continuous Loop Operation
- The system continuously repeats sensing, processing, and control.
- Ensures real-time automatic irrigation without manual intervention.

5.1.7 POWER SUPPLY OPERATION

- The system runs using lithium-ion and HW batteries.
- Provides stable and portable power for field use.

CHAPTER-6
SOFTWARE DEVELOPMENT

6.1 INTRODUCTION TO ARDUINO IDE FOR ESP32

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them. It is available for all operating systems i.e. MAC, Windows, Linux and runs on the Java Platform that comes with inbuilt functions and commands that play a vital role in debugging, editing and compiling the code.

A range of Arduino modules available including Arduino Uno, Arduino Mega, Arduino Leonardo, Arduino Micro and many more. Each of them contains a microcontroller on the board that is actually programmed and accepts the information in the form of code. The main code, also known as a sketch, created on the IDE platform will ultimately generate a Hex File which is then transferred and uploaded in the controller on the board.

The IDE environment mainly contains two basic parts: Editor and Compiler were former is used for writing the required code and later is used for compiling and uploading the code into the given Arduino Module. This environment supports both C and C++ languages.

6.2 INSTALLATION OF ESP32 IN ARDUINO IDE

6.2.1 INTRODUCTION TO ARDUINO IDE FOR ESP32

The Arduino Integrated Development Environment (IDE) is the primary programming tool used in this project. The ESP32 board support is added to the Arduino IDE via the Boards Manager using Espressif's official board package URL. The IDE provides a code editor, compiler, and uploader in a single interface. To program the ESP32 with Arduino IDE: Install the Arduino IDE, add the ESP32 board URL in Preferences, install the ESP32 board package via Boards Manager, install required libraries (ESP32Servo, Adafruit SSD1306, Adafruit GFX), connect the ESP32 via USB, select the appropriate board (ESP32 Dev Module) and COM port, and upload the sketch.

FIG 6.1: USB CABLE USED FOR ESP32

STEP 1: INSTALLING ARDUINO IDE

FIG 6.2: ARDUINO OFFICIAL WEBSITE- DOWNLOAD PAGE

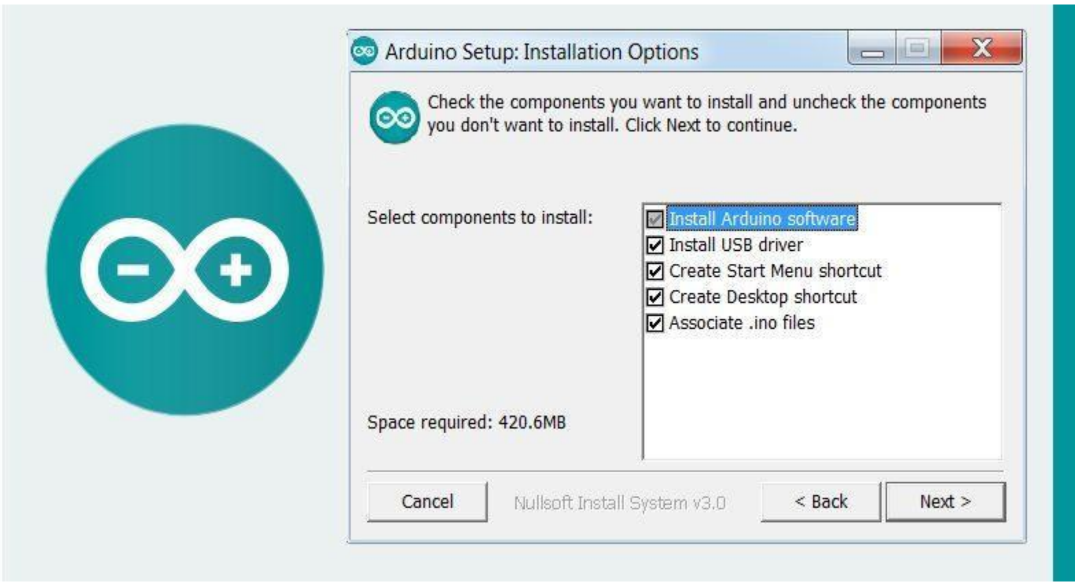

FIG 6.3:ARDUINO IDE DOWNLOAD

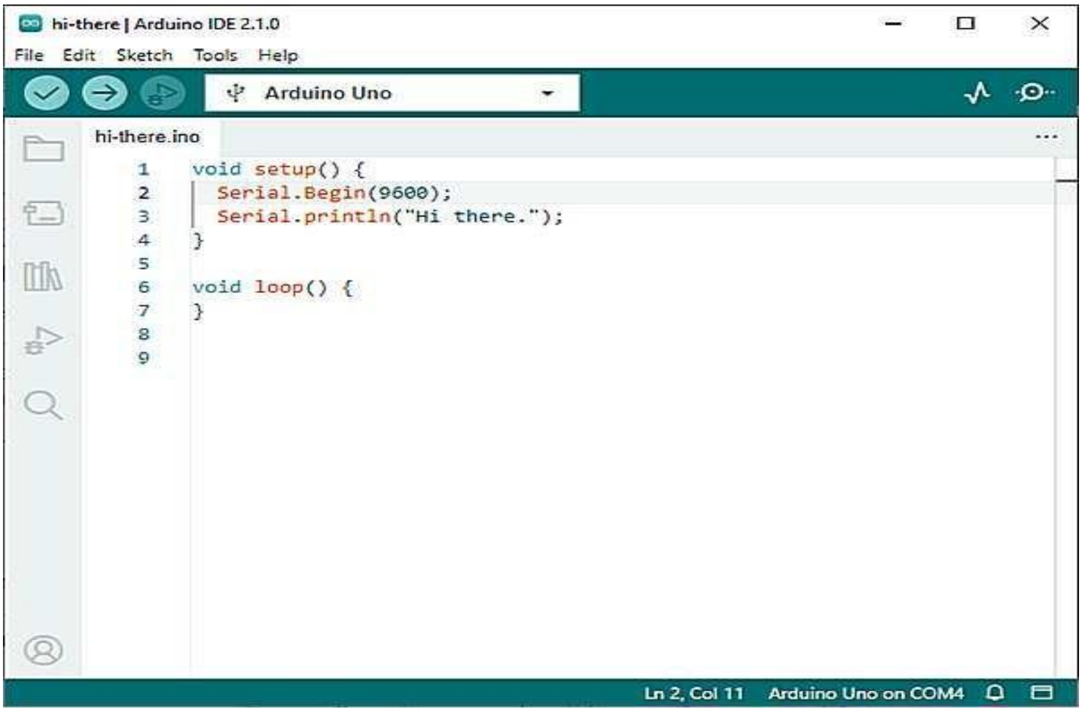

FIG 6.4: ARDUINO IDE INSTALLLATION

Procedure:

1. Go to the Arduino official website (https://www.arduino.cc/en/software)
2. Download the latest version of Arduino IDE for your OS (Windows/Mac/Linux)
3. Run the installer and complete the installation
4. Open Arduino IDE after installation is complete

STEP 2: INSTALLING ESP32 BOARD IN ARDUINO IDE

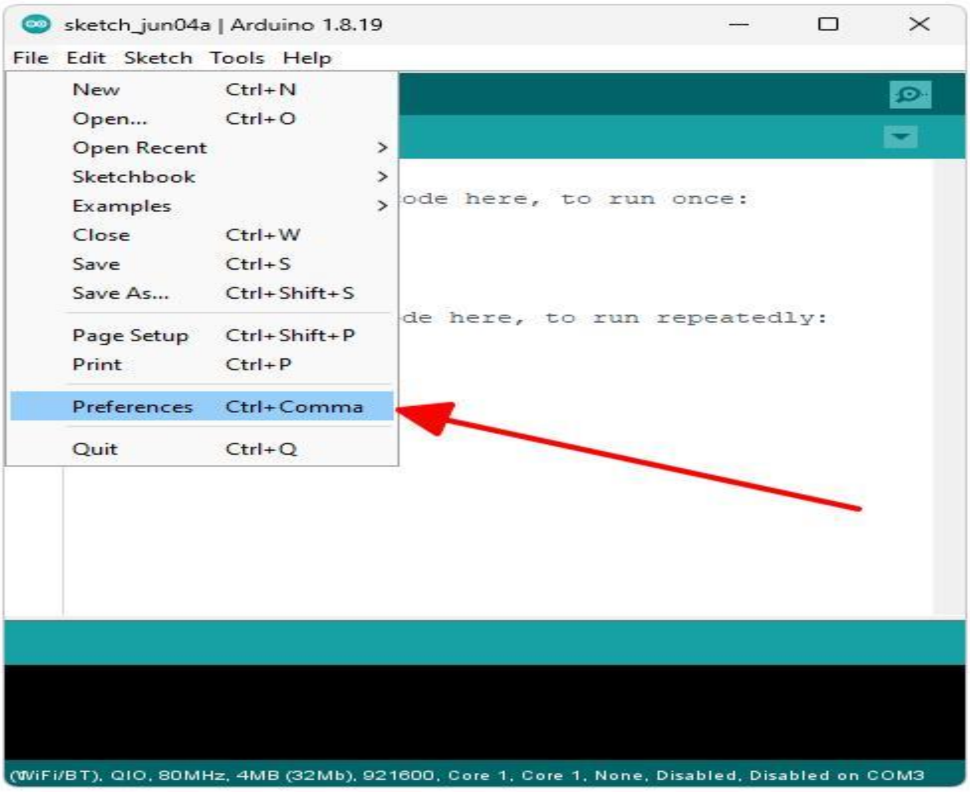

FIG 6.5: BOARD MANAGER -ESP32 SEARCH

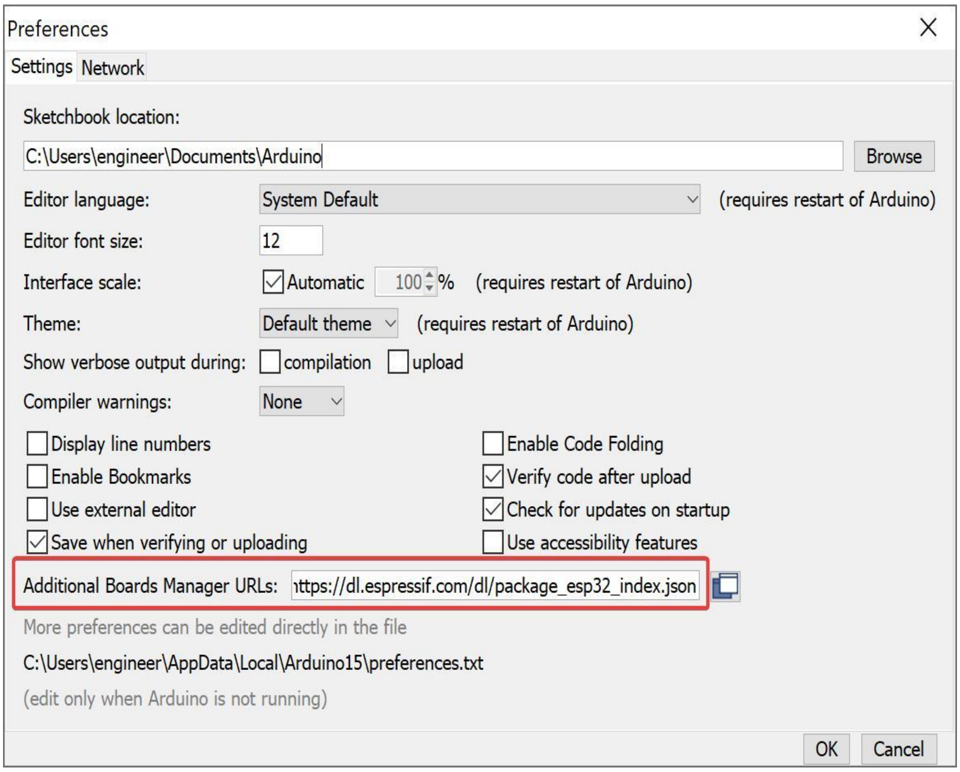

FIG 6.6: ADDING ESP32 BOARD URL in PREFERENCE

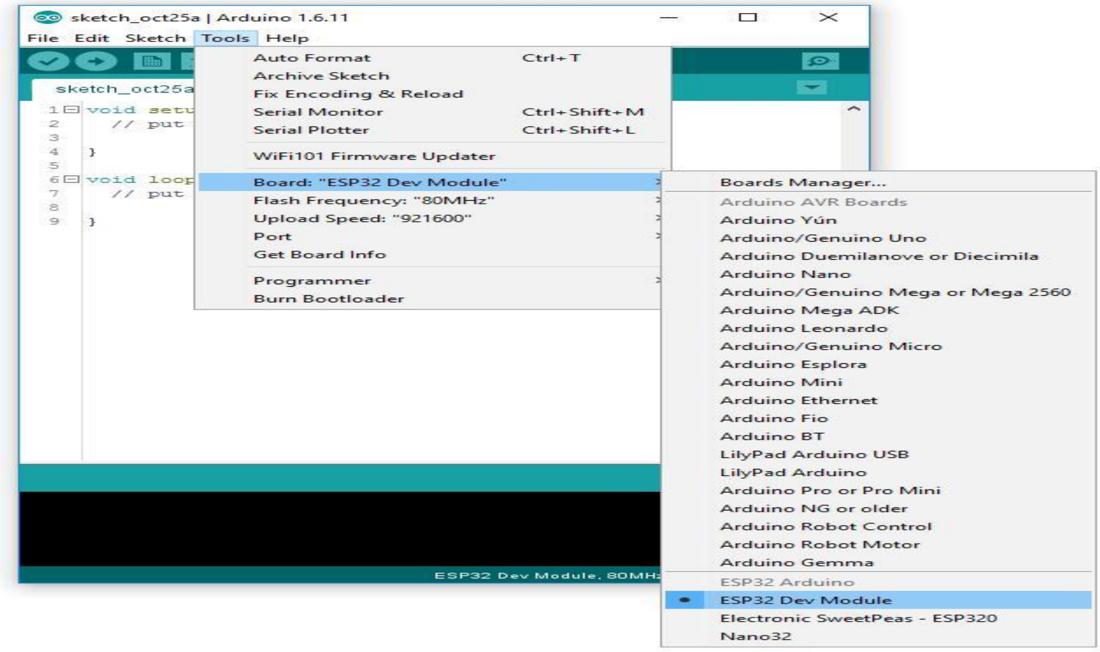

FIG 6.7: INSTALLING ESP32 BOARD PACKAGE.

Steps:

5. Open Arduino IDE.
6. Go to File → Preferences.
7. In “additional board manager URL.s” add:.
8. “Http:// dl.espressif.com/package_esp32_index.json”.
9. Go to Tools → Board → Boards Manager.
10. Search “ESP32” in the search box Open.
11. Select the package by Espressif Systems and click Install.

STEP 3: INSTALLING REQUIRED LIBRARIES

Required Libraries: WiFi, WiFi Client Secure, Universal Telegram Bot, Wire, LiquidCrystal_I2C.

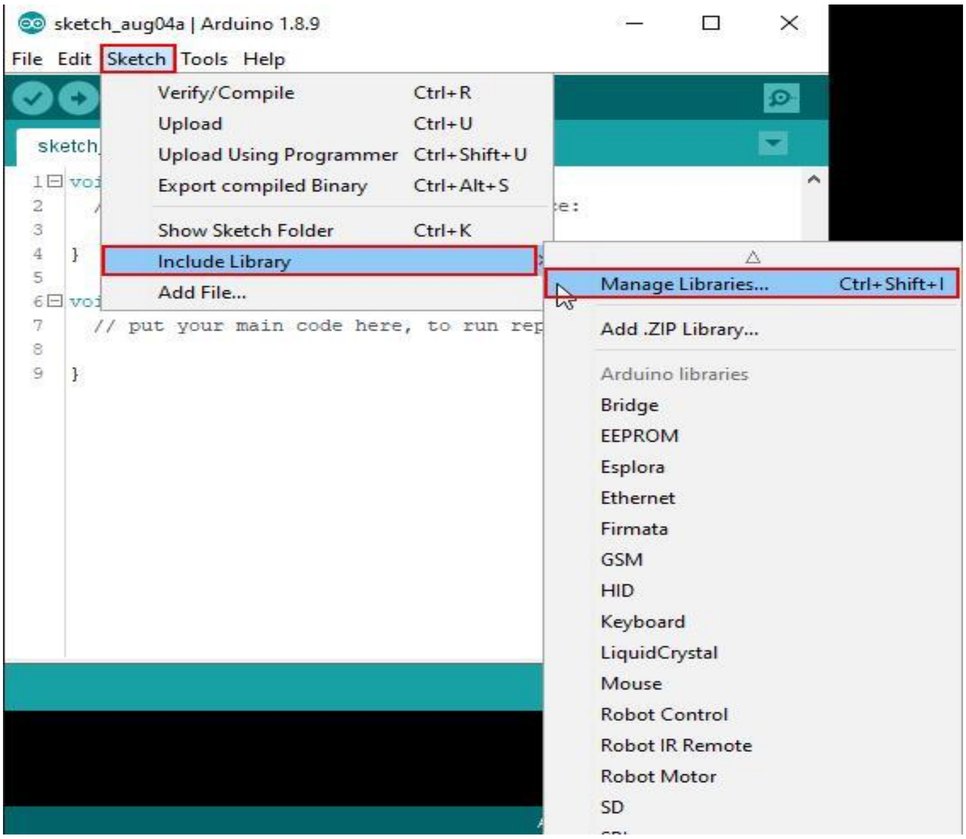

FIG 6.8: LIBRARY MANAGER-SEARCH LIBRARIES

FIG 6.9: INSTALLING LIBRARIES

Steps:

12. Go to Sketch → Include Library → Manage Libraries.
13. Search each library by name in the search bar.
14. Select the correct library and click Install.

STEP 4: WRITING THE CODE

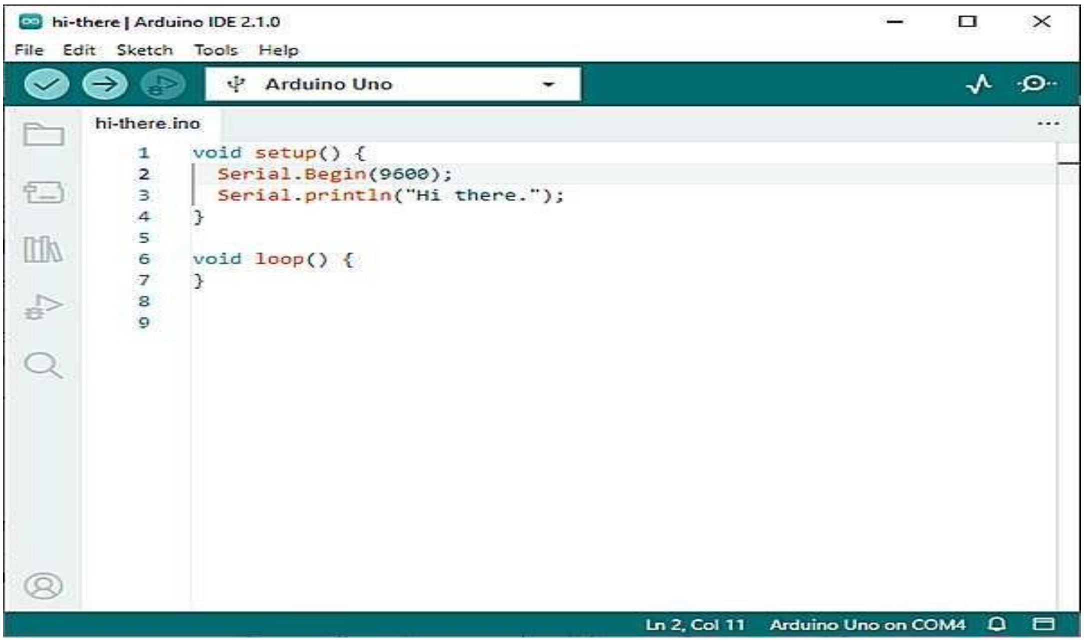

FIG 6.10: ARDUINO IDE CODE EDITOR

Description:

15. Write program in Arduino IDE and include required libraries.
16. Define pins and variables at the top of the sketch.
17. Implement sensor data detection: ultrasonic water level, MQ135 gas sensor, flow sensor.
18. Add water level calculation and threshold monitoring logic.
19. Implement gas leakage detection using MQ135 sensor.
20. Calculate flow rate using flow sensor interrupt.
21. Add alert system using buzzer for local indication.
22. Implement Telegram alert notification via WiFi.
23. Add LCD display code for real-time monitoring.

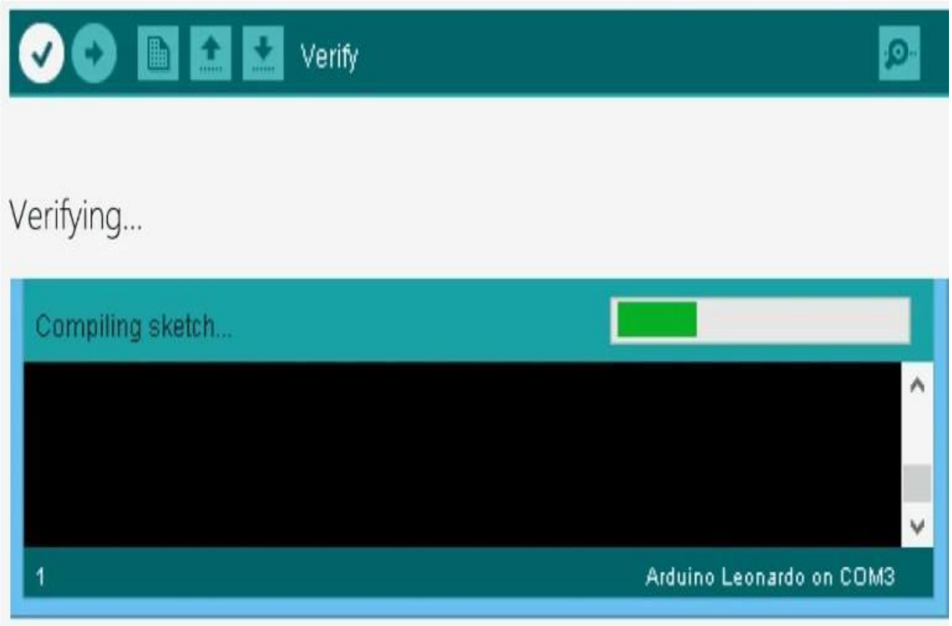

FIG 6.11: COMPILING CODE - VERIFY BUTTON

Steps:

24 Click the Verify button (✓) in Arduino IDE toolbar.

25 Check the output console for any errors.

26 Fix all errors if present before uploading.

STEP 5: COMPILING THE CODE

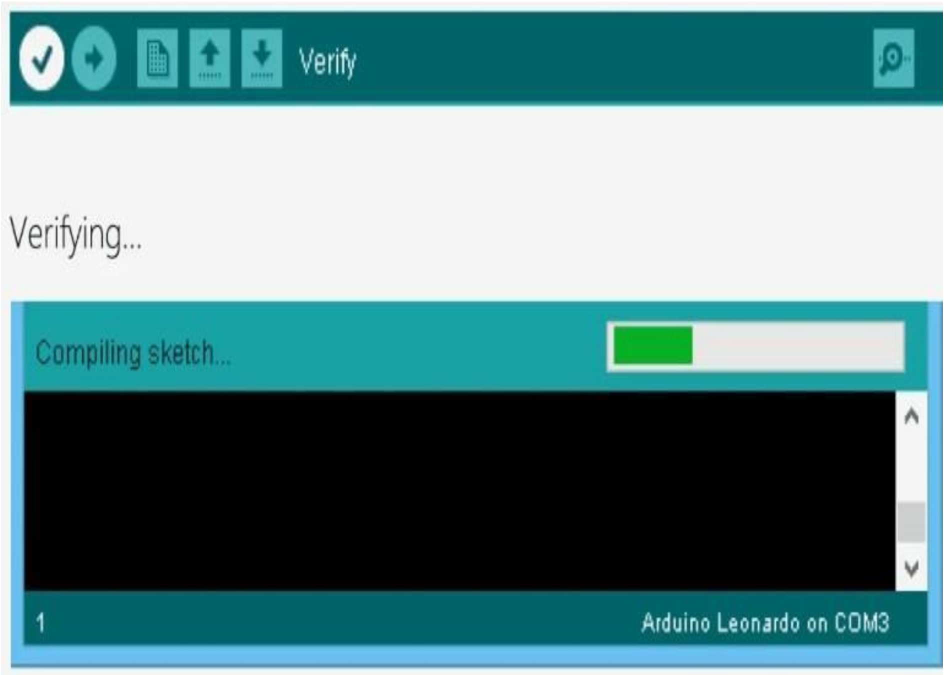

FIG 6.12: COMPILING THE CODE

Steps:

1. Click Verify (✓)
2. Check for errors
3. Fix errors if present

STEP 6: UPLOADING THE CODE TO ESP32

FIG 6. 13: UPLOADING CODE TO ESP32

Steps:

27. connect ESP32 to computer via USB cable.
28. Select Board: Tools → Board → ESP32 Dev Module.
29. Select Port: Tools → Port → COM port (e.g. COM3 on Windows).
30. Click the Upload button (→) to flash the code onto the ESP32.

STEP 7: SERIAL MONITOR

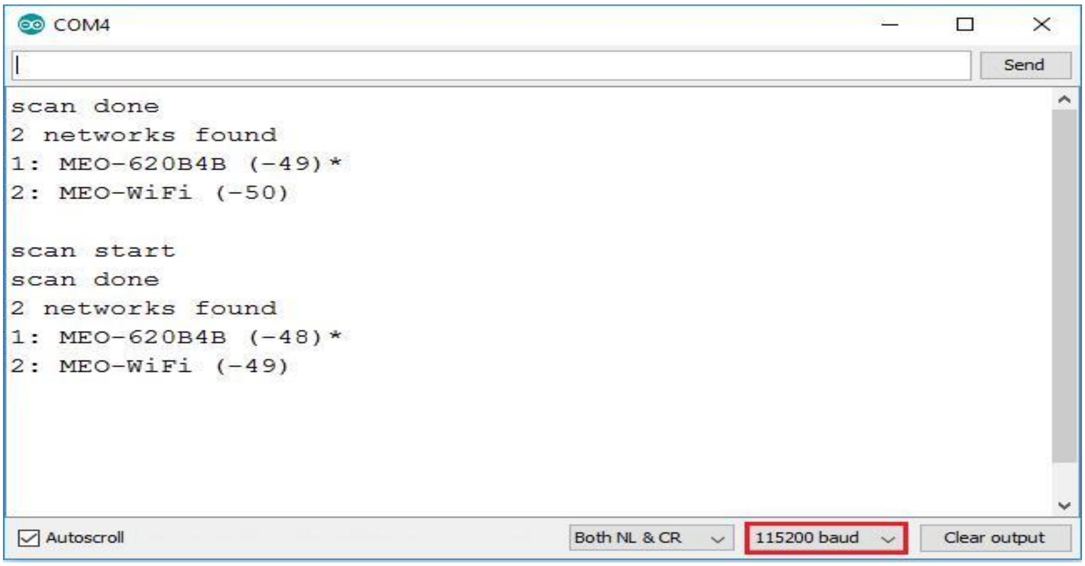

FIG 6.14: SERIAL MONITOR OUTPUT

Steps:

31. Open Serial Monitor from Tools → Serial Monitor (or Ctrl+Shift+M)
32. Set baud rate to 115200 to match the code

Used for: Debugging sensor readings, checking WIFI connection status and monitoring.

STEP 8: SYSTEM EXECUTION FLOW

33. System initializes: ESP32, sensors, LCD display, and WIFI module
34. WIFI connection is established and Telegram bot starts
35. Sensors continuously monitor parameters in real time
36. Water level is measured using ultrasonic sensor
37. Gas level is detected using MQ135 sensor
38. Flow rate is calculated using flow sensor interrupt
39. System checks threshold conditions for water, gas, and flow
40. If abnormal condition detected, buzzer alert is triggered
41. Telegram message is sent via WiFi to the user's phone
42. Real-time data is displayed on the 16x2 LCD screen

STEP 9: TELEGRAM BOT SETUP FOR ALERT NOTIFICATIONS

Install Telegram App:

- Download Telegram from Play Store / App Store and sign in using your mobile number.

Search for BotFather:

- In the Telegram search bar, type BotFather and select the official verified bot (✓ verified).

Start BotFather:

- Click Start, then type the command: /newbot.

Create Your Bot:

- Enter a Bot Name (example: Drainage Alert Bot). Enter a Username that must end with “bot” (example: drainage_alert_bot). After creation you will receive a message with your BOT TOKEN — save it.

Get Chat ID:

- Search your bot in Telegram and press Start. Then open a browser and visit:
- https://api.telegram.org/bot<YOUR_BOT_TOKEN>/getUpdates — you will see your Chat ID as: “chat”:{“id”:123456789}.

Add to ESP32 Code:

- Replace the placeholders in your sketch: #define BOT_TOKEN “your_token_here” and #define CHAT_ID “your chat id”.

Test Notification:

- Upload the code to ESP32. When an alert condition occurs (water overflow, gas leak, or flow blockage), you will receive a Telegram notification message on your mobile device.

CHAPTER-7
RESULTS

7.1 RESULTS:

The results of the Automatic Smart Irrigation and Monitoring System using ESP32 typically include the following aspects:

Soil Moisture Monitoring Accuracy:

- The system's ability to accurately measure soil moisture levels using the sensor.
- Proper detection of dry and wet soil conditions.
- Performance under different soil types and environmental conditions.

Automatic Irrigation Control:

- Automatic switching ON/OFF of the water pump based on soil moisture levels.
- Reduction of manual intervention in irrigation.
- Efficient water usage by supplying only required amount of water.

Water Flow Monitoring:

- Accurate measurement of water flow using flow sensor.
- Monitoring total water usage for irrigation.
- Detection of abnormal flow conditions (e.g., leakage or blockage).

Real-time Monitoring and Display:

- Display of soil moisture and system status on LCD.
- Real-time updates of irrigation process.
- Easy understanding of system operation for users.

System Reliability:

- Continuous operation without system failure.
- Stable performance of sensors and ESP32 controller.
- Proper functioning even under varying environmental conditions.

Energy Efficiency:

- Efficient power usage using low-power components.
- Battery backup support using Lithium-ion battery.
- Reduced energy consumption during idle conditions.

Cost Effectiveness:

- Low-cost components used in the system.

- Affordable solution for farmers and small-scale users.
- Reduction in water and electricity costs over time.

FIG :7.1 INITALAIZATION OF SMART AGRO

The LCD display shows the message "SMART AGRO Starting…", indicating system startup.It confirms that the microcontroller (ESP32) is powered ON and functioning properly.During this stage, all connected components like soil moisture sensor, water flow sensor, relay,and pump are initialized.The system performs a self-check to ensure sensors and modules are working correctly.

FIG :7.2 OUTPUT DISPLAY OF DRY SOIL

- The LCD displays real-time sensor readings of the system.
- SM: 0% indicates that the soil moisture level is very low (dry condition).

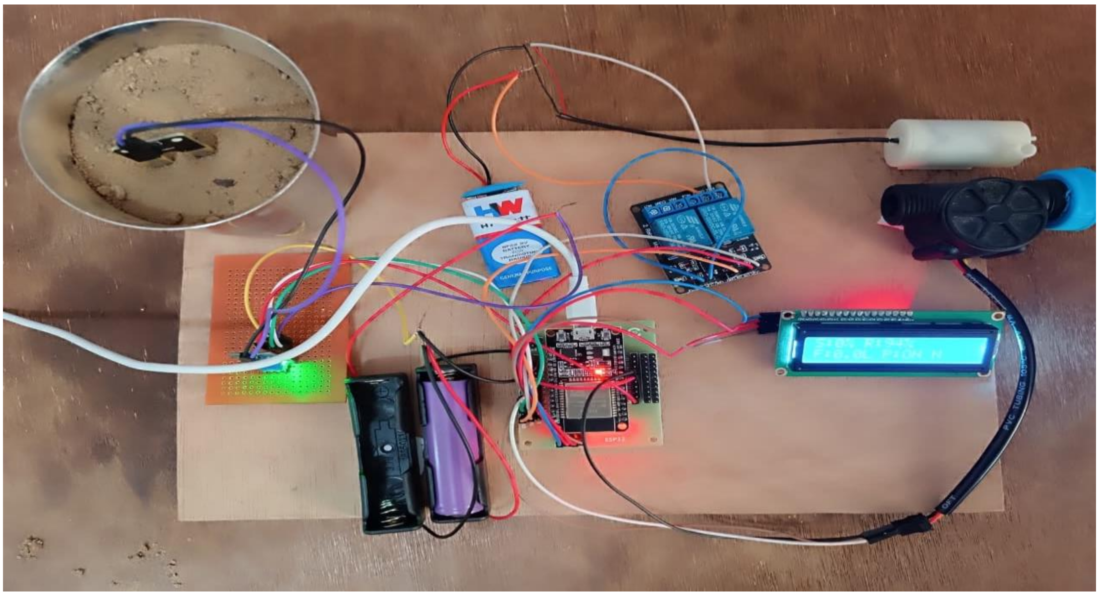

FIG:7.3 DETECTION OF DRY SOIL

The fig 7.3 says designed system successfully detects the moisture level of the soil and identifies dry conditions effectively. When the soil moisture falls below the predefined threshold, the sensor detects the dryness and sends a signal to the microcontroller. Based on this input, the system activates the output components such as the relay.

FIG 7.4: OUTPUT DISPLAY OF WET SOIL

The above figure 7.4 shows the real-time output displayed on the LCD screen when the soil is in a wet condition. The system continuously monitors soil moisture levels using a soil moisture sensor and processes the data through the microcontroller.

In this condition, the moisture level is high (indicated as S: 66%), which means the soil already contains sufficient water. Therefore, the irrigation system automatically turns OFF the water pump (P: OFF) to prevent overwatering.

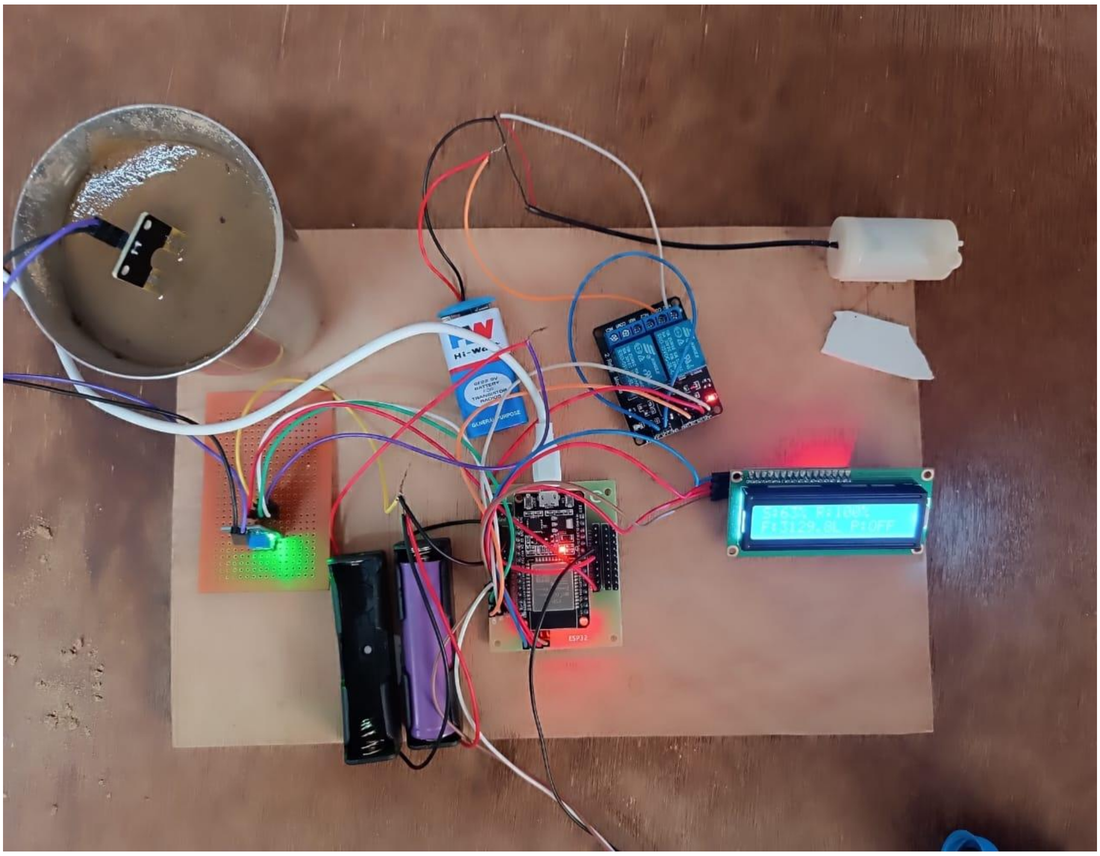

FIG:7.5 DETECTION OF WET SOIL

The above figure7.5 shows the experimental setup of the Automatic Smart Irrigation and Monitoring System during the detection of wet soil. In this setup, the soil moisture sensor is placed inside the soil to continuously measure the moisture level

When the soil contains sufficient water, the sensor detects a high moisture value and sends this data to the microcontroller. Based on this input, the controller processes the signal and determines that irrigation is not required.

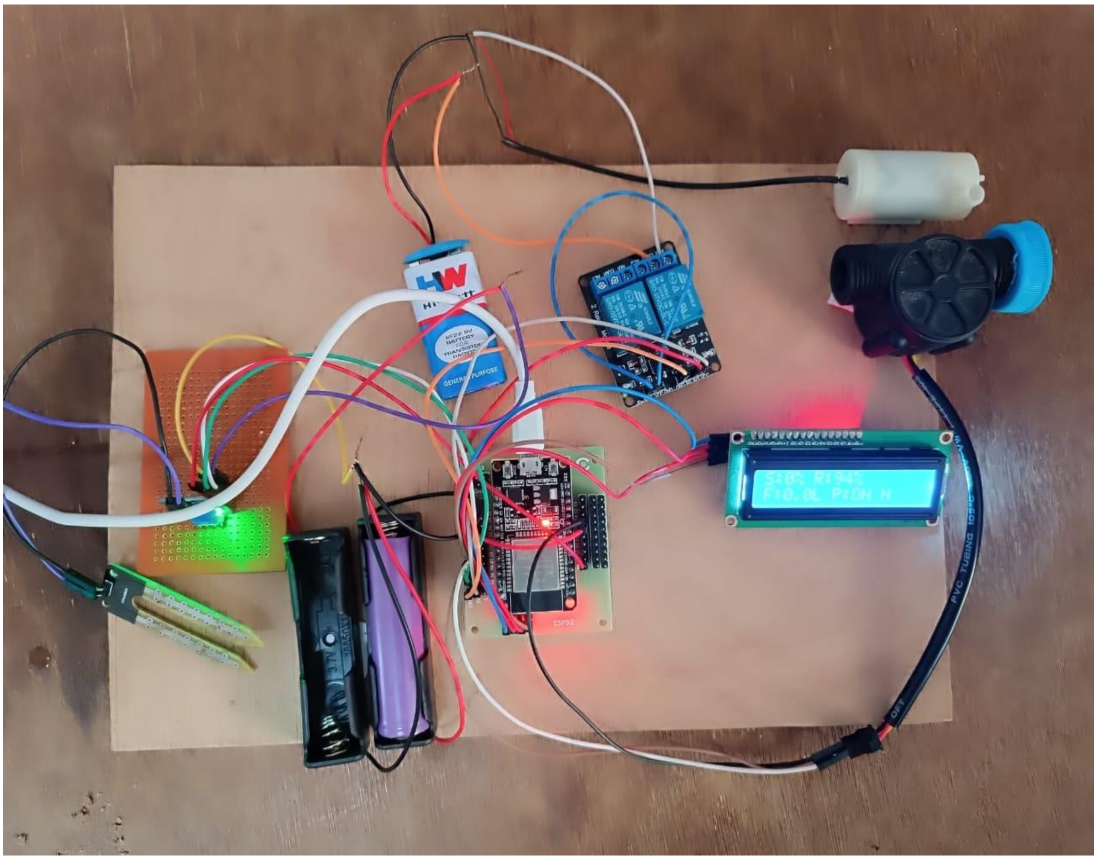

FIG:7.6 FINAL PROTOTYPE OF PROPOSED METHOD

The above figure7.6 shows the final prototype of the Automatic Smart Irrigation and Monitoring System. This prototype integrates all the hardware components required to automatically monitor soil moisture and control irrigation efficiently.

The system consists of a soil moisture sensor, microcontroller (Arduino/ESP module), relay module, water pump, LCD display, and a power supply unit. All components are connected and mounted on a base board to demonstrate the real-time working of the system

CHAPTER-8
ADVANTAGES AND APPLICATIONS

8.1 ADVANTAGES:

Accuracy and Efficiency:

- Accurate Monitoring: Soil moisture sensors provide precise measurement of soil conditions.
- Automated Process: The system automatically controls irrigation, reducing manual effort.
- Time Saving: Farmers do not need to continuously monitor the field.

Water Conservation:

- Efficient Usage: Water is supplied only when required based on soil moisture levels.
- Prevents Overwatering: Reduces water wastage and protects crops from damage.
- Optimized Irrigation: Ensures proper distribution of water to plants.

Cost-Effective and Scalable:

- Low Cost: Uses affordable components like ESP32 and sensors.
- Reduced Labor Cost: Minimizes need for human intervention.
- Scalable: Can be expanded for large agricultural fields.

Real-Time Monitoring:

- Live Data Display: Displays soil moisture and system status on LCD.
- Quick Response: Immediate action when soil becomes dry.
- Better Decision Making: Helps users monitor and manage irrigation effectively.

Energy Efficiency:

- Low Power Consumption: Uses energy-efficient components.
- Battery Support: Can operate using Lithium-ion battery backup.
- Smart Operation: System works only when needed, saving energy.

Improved Crop Yield:

- Healthy Plant Growth: Maintains optimal soil moisture levels.
- Prevents Crop Damage: Avoids under-irrigation and over-irrigation.
- Better Productivity: Leads to increased agricultural output.

8.2 DISADVANTAGES:

- High Initial Cost: Expensive setup of sensors, controllers, and infrastructure like wiring, internet, and power supply.

- Technical Complexity: Requires skilled knowledge for operation, maintenance, and proper sensor configuration.
- Internet & Power Dependence: Needs stable internet and continuous power supply for smooth functioning.

8.3 APPLICATIONS:

Agriculture Fields:

- Used in farms to automatically irrigate crops based on soil moisture levels.
- Helps farmers in efficient water management.
- Improves crop productivity and reduces manual work.

Gardening:

- Suitable for home gardens and lawns for automatic watering.
- Maintains proper moisture for plants without daily monitoring.
- Useful for busy individuals and urban gardening.

Greenhouses:

- Maintains controlled environment for plant growth.
- Ensures consistent watering inside greenhouses.
- Supports advanced farming techniques.

Smart Cities:

- Used in public parks and roadside plantations.
- Helps in efficient water usage in urban areas.
- Supports sustainable city development.

Nurseries:

- Used in plant nurseries for proper plant care.
- Provides uniform irrigation to all plants.
- Reduces human effort in watering plants.

Research and Development:

- Used in agricultural research for soil and water analysis.
- Helps in testing new irrigation techniques.
- Supports innovation in smart farming technologies.

CHAPTER-9

CONCLUSION & FUTURE SCOPE

9.1 CONCLUSION:

The Automatic Smart Irrigation and Monitoring System efficiently automates the irrigation process using sensors and a microcontroller. It supplies water based on soil moisture levels, reducing water wastage and manual effort. The system is cost-effective, reliable, and suitable for agricultural use, especially in areas without internet access. Overall, it enhances crop productivity and supports sustainable farming practices.

9.2 FUTURE SCOPE:

The Automatic Smart Irrigation and Monitoring System can be further enhanced by integrating advanced technologies to improve efficiency and usability. It can be upgraded with IoT connectivity or remote monitoring and control through mobile applications. Incorporating weather forecasting can help optimize irrigation schedules based on rainfall and climate conditions. The use of solar power can make the system more energy-efficient and eco-friendly. Additionally, AI-based data analysis can be implemented to predict crop water requirements more accurately, making the system smarter and more adaptive for modern agriculture.

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APPENDIX

SOURCE CODE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
// LCD
LiquidCrystal_I2C lcd(0x27, 16, 2);
// Pins
#define SOIL_PIN 34
#define RAIN_PIN 35
#define FLOW_PIN 27
#define RELAY_PIN 26
// Thresholds (adjust if needed)
int soilThreshold = 1800;
int rainThreshold = 1500;
// Flow variables
volatile int pulseCount = 0;
float flowRate = 0;
unsigned long lastTime = 0;
// Interrupt for flow sensor
void IRAM_ATTR flowPulse() {
  pulseCount++;
}
void setup() {
  Serial.begin(115200);
  // LCD
  Wire.begin(21, 22);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0,0);
  lcd.print("SMART AGRO");
```

```
  lcd.setCursor(0,1);
  lcd.print("Starting...");
  delay(2000);
  lcd.clear();
  // Pins
  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, HIGH); // Pump OFF
  pinMode(FLOW_PIN, INPUT_PULLUP);
  attachInterrupt(digitalPinToInterrupt(FLOW_PIN), flowPulse, RISING);
}
void loop() {
  // Read sensors
  int soilValue = analogRead(SOIL_PIN);
  int rainValue = analogRead(RAIN_PIN);
  // Convert to percentage
  int soilPercent = map(soilValue, 4095, 0, 0, 100);
  int rainPercent = map(rainValue, 4095, 0, 0, 100);
  // Flow calculation (every 1 second)
  if (millis() - lastTime >= 1000) {
    flowRate = pulseCount * 2.25;
    pulseCount = 0;
    lastTime = millis();
  }
  // Pump logic
  if (soilValue < soilThreshold && rainValue < rainThreshold) {
    digitalWrite(RELAY_PIN, LOW);   // ON
  } else {
    digitalWrite(RELAY_PIN, HIGH);  // OFF
  }
  // -------- LCD DISPLAY --------
```

```
  lcd.setCursor(0,0);
  lcd.print("S:");
  lcd.print(soilPercent);
  lcd.print("% ");
  lcd.print("R:");
  lcd.print(rainPercent);
  lcd.print("%   ");  // clear
  lcd.setCursor(0,1);
  lcd.print("F:");
  lcd.print(flowRate,1);
  lcd.print("L ");
  if (digitalRead(RELAY_PIN) == LOW)
    lcd.print("P:ON ");
  else
    lcd.print("P:OFF");
  // -------- SERIAL --------
  Serial.println("------ STATUS ------");
  Serial.print("Soil: "); Serial.println(soilValue);
  Serial.print("Rain: "); Serial.println(rainValue);
  Serial.print("Flow: "); Serial.println(flowRate);
  // -------- ALERT --------
  if (digitalRead(RELAY_PIN) == LOW && flowRate < 1) {
    Serial.println("ALERT: No water flow!");
  }
  delay(1000);
}
```